

WASP-3b

R-0287

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_170601 · created 2026-07-17T17:14:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457905.8587 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.033
Transit depth (Rp/Rs)^2	2.03 +/- 0.95 [%]
Orbital Inclination (inc)	84.35 +/- 2.67
Airmass coefficient 1 (a1)	1.133 +/- 0.032
Airmass coefficient 2 (a2)	-0.111 +/- 0.057
Scatter in the residuals of the lightcurve fit is	2.83 %
Best Comparison Star	#3 - [275, 195]
Optimal Aperture	2.69
Optimal Annulus	9.51
Transit Duration (day)	0.118 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.033 vs 0.10538 (deviation 1.07σ)
Duration fit vs published	0.118 d vs 0.1167 d (deviation 0.08σ)
Mid-time O–C	6.3 min
Depth SNR	4.1
Residual scatter	2.83 %

Light curve

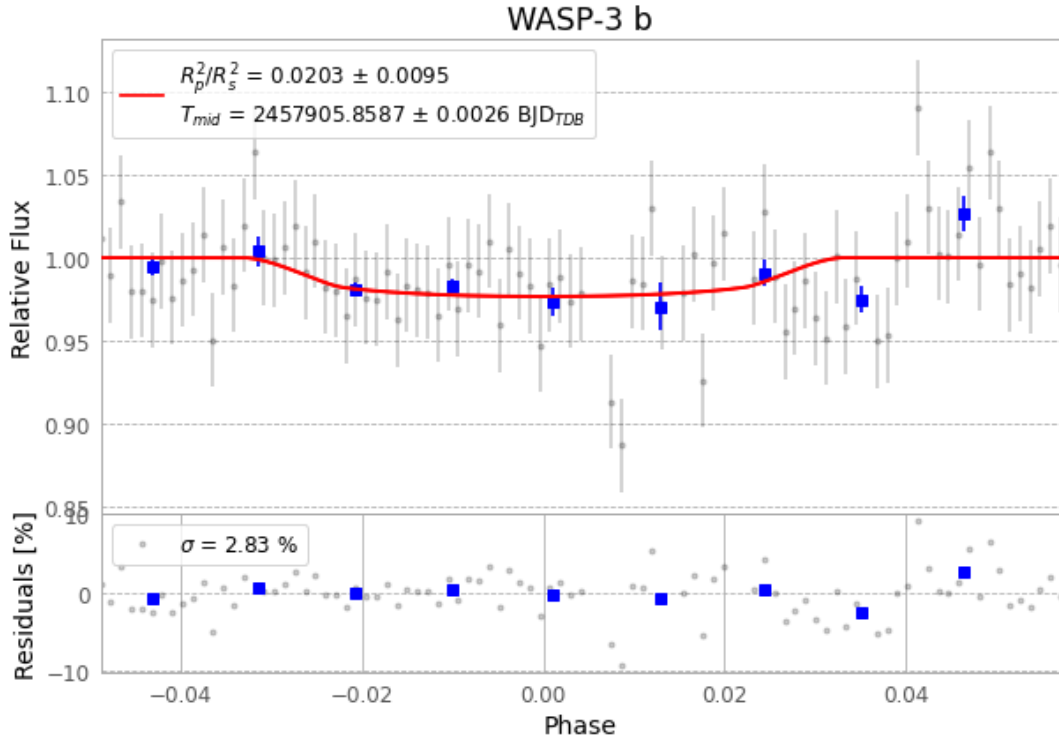


Figure 1 — FinalLightCurve_WASP-3 b_2017-05-31.png (SHA-256 782ff4e9d9ac1b91...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 98cfaa4899d27d0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

95 FITS frames (62 MB), WASP-3170601062112.FITS → WASP-3170601110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-170601.log (6124db530030...), FinalLightCurve_WASP-3 b_2017-05-31.png (782ff4e9d9ac...), FinalLightCurve_WASP-3 b_2017-05-31.csv (bae4866c3fff...)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 17:14Z.